

The Boundary Trace Algebra of Monogenic Spinors Need Not Be Isomorphic

Automated Math Discovery Candidate Counterexample

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Status. Candidate full negative answer to open item (ii) in the further-results section of arXiv:2203.00118, likely valid pending expert review. On the Euclidean unit three-ball, the algebra generated by monogenic spinor fields contains the nonzero scalar field $1 - |x|^2$, whose boundary trace is zero.

Source Question

Roberts asks whether the algebra generated by monogenic spinor fields and the algebra generated by their boundary traces are isometrically isomorphic under the trace map [1]. The question is item (ii) on page 22:

$f|_{\partial M}$. In essence, the Hilbert transform yields boundary values of functions conjugate in the generalized Cauchy–Riemann equations given by $\nabla f = 0$. In fact, if ϕ_k is a k -vector, the Hilbert transform of ϕ_k contains a $k - 2$ -, a k -, and $k + 2$ -vector component. This is part of why we suggest to look beyond pairwise conjugate forms.

A reasonable question to ask is: are these two Hilbert transform operators related in some way? Moreover, Sharafutdinov and Shonkwiler extend Belishev and Sharafutdinov’s Dirichlet-to-Neumann operator to the complete Dirichlet-to-Neumann operator [8]. If it is not possible to relate Belishev and Sharafutdinov’s Hilbert transform using the DN operator on forms to the Hilbert transform in Clifford analysis, is there a related operator defined in terms of the complete DN operator that relates to Clifford analysis?

- ii. This point is essentially given as an open question by Belishev and Vakulenko [13]. Specifically, those two ask whether it is true that the algebras $\vee \mathcal{M}^+(M)$ and $\vee \text{tr} \mathcal{M}^+(M)$ are isometrically isomorphic. At the moment, we have a partial answer: by the Cauchy integral formula, a monogenic field $f \in \mathcal{M}(M)$

Here $\vee \mathcal{M}^+(M)$ denotes the pointwise algebra generated by the monogenic even Clifford-valued fields. We show that, already for the unit ball B^3 , boundary restriction on this algebra has a nontrivial kernel.

Answer and Proof Intuition

Theorem 1. *Let $B^3 = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : |x| \leq 1\}$, equipped with its Euclidean metric and positive Clifford convention $e_i^2 = 1$. The boundary trace homomorphism*

$$\text{tr} : \vee \mathcal{M}^+(B^3) \longrightarrow \vee \text{tr} \mathcal{M}^+(B^3)$$

is not injective. In particular, it is neither an algebra isomorphism nor an isometric isomorphism.

The point is that products of monogenic fields need not remain monogenic, and the generated algebra is consequently much larger than the monogenic space itself. Two linear monogenic coordinates already generate the radial quadratic $|x|^2$. Subtracting it from the constant field gives an element which vanishes on the sphere but not in the ball.

Proof

Work in the even Clifford algebra Cl_3^+ . Put

$$I = e_1 e_2, \quad J = e_1 e_3, \quad K = e_2 e_3.$$

Then

$$I^2 = J^2 = K^2 = -1, \quad IJ = -K, \quad IK = J, \quad JK = -I,$$

with the reversed products having the opposite signs. Define

$$Z = x_2 - x_1 I, \quad W = x_3 - x_1 J.$$

For the Euclidean Hodge–Dirac operator $D = e_1 \partial_{x_1} + e_2 \partial_{x_2} + e_3 \partial_{x_3}$,

$$DZ = e_2 - e_1(e_1 e_2) = 0, \quad DW = e_3 - e_1(e_1 e_3) = 0.$$

Thus $Z, W \in \mathcal{M}^+(B^3)$. Right multiplication by a constant even Clifford element preserves monogenicity, so all factors below belong to the generating space used to form $\bigvee \mathcal{M}^+(B^3)$.

Direct Clifford multiplication gives the identity

$$\begin{aligned} x_1^2 + x_2^2 + x_3^2 &= Z^2 + \frac{1}{2} ZWK + ZIWJ \\ &\quad + \frac{1}{2} WZK - WIWI. \end{aligned} \tag{1}$$

For completeness, the five terms on the right have components in the ordered basis $(1, I, J, K)$

$$\begin{aligned} Z^2 &= (-x_1^2 + x_2^2, -2x_1 x_2, 0, 0), \\ \frac{1}{2} ZWK &= (\frac{1}{2} x_1^2, \frac{1}{2} x_1 x_2, -\frac{1}{2} x_1 x_3, \frac{1}{2} x_2 x_3), \\ ZIWJ &= (x_1^2, x_1 x_2, x_1 x_3, -x_2 x_3), \\ \frac{1}{2} WZK &= (-\frac{1}{2} x_1^2, \frac{1}{2} x_1 x_2, -\frac{1}{2} x_1 x_3, \frac{1}{2} x_2 x_3), \\ -WIWI &= (x_1^2 + x_3^2, 0, 0, 0). \end{aligned}$$

Their non-scalar components cancel, and their scalar components sum to $|x|^2$, proving (1). Every term in its right-hand side is a finite sum or product of elements of $\mathcal{M}^+(B^3)$; hence $|x|^2 \in \bigvee \mathcal{M}^+(B^3)$.

The constant field 1 is monogenic, so

$$P(x) = 1 - |x|^2$$

also lies in $\bigvee \mathcal{M}^+(B^3)$. Yet $P|_{\partial B^3} = 0$, while $P(0) = 1$. Therefore $P \neq 0$ belongs to the kernel of the boundary trace homomorphism. This proves Theorem 1. $\square$

Scope, Checks, and Review Focus

The counterexample addresses exactly the generated-algebra question, not the paper’s separate question about extending its spinor-spectrum theorem to all compact Riemannian manifolds. It also explains why the maximum principle for individual monogenic fields does not pass to their generated algebra: the kernel element $1 - |x|^2$ is not itself monogenic.

The identity (1) was derived by an exact rational linear-algebra search in the degree-two Clifford polynomial space and then checked by an independent symbolic expansion retained with this packet. A bounded novelty search through August 11, 2026 used the exact paper title, the wording of the trace-algebra question, and combinations of “monogenic”, “boundary trace”, and “isometrically isomorphic”; no later primary source giving this negative answer was found. Expert review should focus on matching the paper’s $\sqrt{}$ -notation with the generated pointwise algebra and on the Clifford sign convention in (1).

References

- [1] C. Roberts, *A Gelfand Transform for Spinor Fields on Embedded Riemannian Manifolds*, arXiv:2203.00118v1 (2022), item (ii), p. 22.
- [2] M. I. Belishev and A. F. Vakulenko, *On algebraic and uniqueness properties of 3d harmonic quaternion fields*, arXiv:1901.09201 (2019).